

A Collection of Numerical Patterns and Processes with Several Computed Visuals

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1 Binary Sierpinski, Singular Values

Take a matrix, B , filled with the row-vectors of consecutive binary numbers of length k (beginning with 0 and up to 2^k), and compute;

$$BB^T$$

This will generate an exhaustive array of the dot-product of digits between all 2^k binary strings and every other one; an effect which leaves a specific, familiar pattern of zeros;

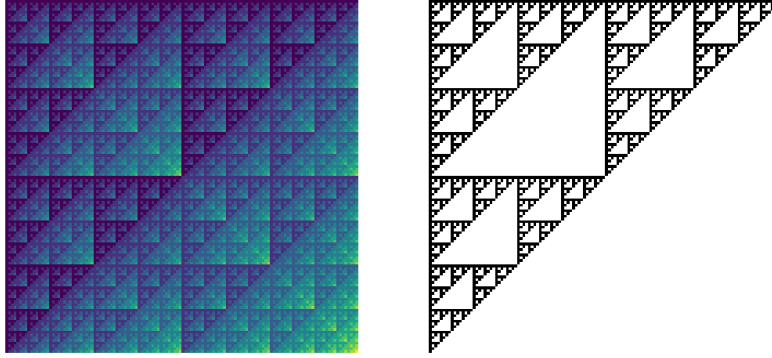


Figure 1: $k = 7$ Binary Triangle Construction

Of note also are the predictable singular values of B ; (to be addressed shortly).

$$\sqrt{(k+1)2^{k-2}}, \sqrt{2^{k-2}}, \sqrt{2^{k-2}}, \dots$$

As well studied as the Sierpinski triangle is, it is not surprising that binary constructions such as this are quite common. In this case, the recursive nature of block matrix multiplication is to blame;

$$B_k B_k^T = \begin{pmatrix} 0 & B_{k-1} \\ 1 & B_{k-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ B_{k-1}^T & B_{k-1}^T \end{pmatrix} = \begin{pmatrix} B_{k-1} B_{k-1}^T & B_{k-1} B_{k-1}^T \\ B_{k-1} B_{k-1}^T & 11^T + B_{k-1} B_{k-1}^T \end{pmatrix}$$

Giving three identical matrices and a corner without the possibility of being 0. Checking any small case, such as B_2 , to confirm upper-triangularity completes the construction.

Humorously of course, thinking about the Sierpinski triangle as a matrix (black=0, white=1), and stripping away the outer row and column of 0 gives a determinant of not just a triangular matrix, but an upper Sierpinski triangular one, containing -1 , call it R , on a background of 0;

$$\det(R + 11^T) = \det(R) + 1 \operatorname{adj}(R) 1^T = \sum_{i,j} \det(R_{ji}) (-1)^{(i+j)} = \det(R_{2^k 2^k}) = \det(R_{\operatorname{diag}=-1}) = (-1)^{2^k-1} = -1; \forall k$$

The general case for $B \in \mathcal{R}^{b^k \times k}$ containing row-vectors of the consecutive integers in base b of length k pans out similarly with fractal protrusions, but the zero-regions do not connect together along the diagonal and make up an increasingly smaller percentage of the image.

Fact: The sum of the digits in B in base b is precisely $b^k \frac{(b-1)}{2} k$

On the hunt for more aesthetic matrix products, as well as for the singular values of B , we examine $B^T B$ in base b to find an interesting, yet significantly simpler structure;

$$B_k^T B_k = \begin{pmatrix} \vec{0}^T & \vec{1}^T & \dots & (b-1)^T \\ B_{k-1}^T & B_{k-1}^T & \dots & B_{k-1}^T \end{pmatrix} \begin{pmatrix} \vec{0} & B_{k-1} \\ \vec{1} & B_{k-1} \\ \vdots & \vdots \\ (b-1) & B_{k-1} \end{pmatrix} = \begin{pmatrix} \sum_{j=0}^{b-1} \vec{j}^T \vec{j} & \sum_{j=0}^{b-1} B_{k-1}^T \vec{j} \\ \sum_{j=0}^{b-1} B_{k-1}^T \vec{j} & b B_{k-1}^T B_{k-1} \end{pmatrix}$$

With diagonal entries of;

$$\vec{0} \cdot \vec{0} + \vec{1} \cdot \vec{1} + \dots + (b-1) \cdot (b-1) = b^{k-1} (0^2 + 1^2 + \dots + (b-1)^2) = b^{(k-1)} \frac{b(b-1)(2b-1)}{6}$$

Because each row of B^T has an equal frequency of each digit, the off-diagonal entries become uniform vectors with value;

$$\frac{b(b-1)}{2} B_{k-1}^T \vec{1} = \frac{(b-1)^2}{4} b^k$$

While BB^T gives an infamously intricate gasket, BB^T gives nothing more than a diagonal matrix plus a constant background. Completing the investigation we take the determinant lemma;

$$\det \left\{ \left(\frac{b^{(k-1)} b(b-1)(2b-1)}{6} - \frac{(b-1)^2}{4} b^k - \lambda \right) I + \frac{(b-1)^2}{4} b^k \mathbf{1}\mathbf{1}^T \right\}$$

And find the sequence of singular values for the matrix of consecutive integers of length k in base b as;

$$\sqrt{b^k (b-1) \left(\frac{(2b-1)}{6} + \frac{(b-1)(k-1)}{4} \right)}, \sqrt{b^k \frac{(b-1)(b+1)}{12}}, \sqrt{b^k \frac{(b-1)(b+1)}{12}} \dots$$

2 Chaotic Powers

x^n for $x \in \mathcal{R}$ grows uniformly, while z^n for $z \in \mathcal{C}$ grows in regular, circular, cycles: How can we engineer a 3D coordinate system with truly irregular power behavior?

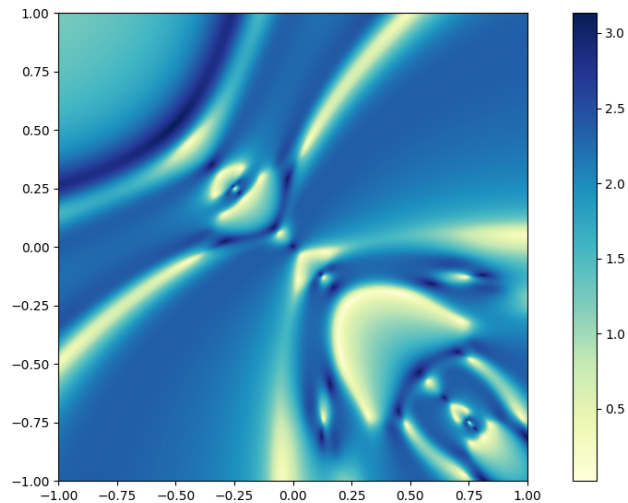
Defining a basis with, $[\hat{x}, \hat{y}, \hat{z}]$, using $\hat{x}^2 = -\hat{x}, \hat{y}^2 = -\hat{y}, \hat{z}^2 = -\hat{z}$ along with a *non-alternating* cross product, $\hat{x}\hat{y} = \hat{z}, \hat{y}\hat{z} = \hat{x}, \hat{x}\hat{z} = \hat{y}$, gives a multiplication pattern with very erratic behavior. Multiplication simply follows this multiplication table for the basis vectors, which can be thought of as covering the 3×3 matrix of all possible pairs of multiplied cross-terms with an assigned value *symmetrically* in each variable.

$$\vec{a} \hat{\times} \vec{b} = \hat{x} (X \cdot ab^T) + \hat{y} (Y \cdot ab^T) + \hat{z} (Z \cdot ab^T)$$

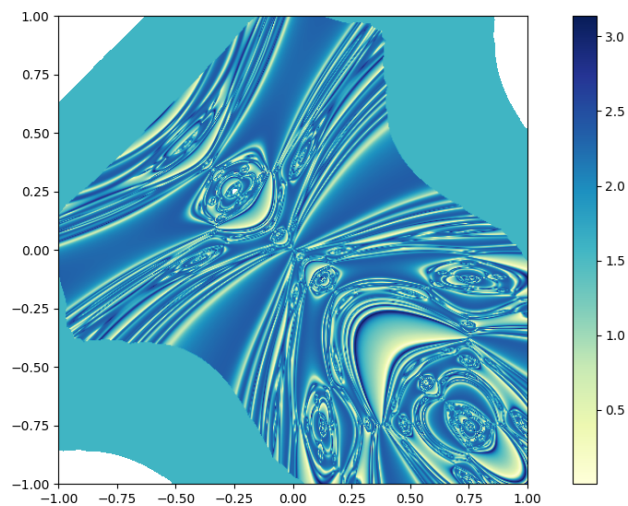
Where $\cdot$ implies the sum of every term-wise multiplication, and where the symmetric matrices X, Y, Z are;

$$X = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

In the figure below, we examine the quantity $(x\hat{x} + y\hat{y} + z\hat{z})^n$ for values in the xy plane, keeping $z = 0.75$ for now, with $n = 5$. Since the magnitude of these quantities can grow quite large, we plot the angle made to the z -axis in spherical coordinate fashion.



Like any good numerical pattern, heightening the iterations ($n = 10$ in the following plot), the better the detail;



A fascinating portrait showing swirling orbits in $\mathcal{R}^3$ and fractal pockets with rapidly diverging values.

2.1 Opportunistic Notation

Taking a small aside, the idea of defining multiplication as a *covering* of the matrix of coefficients ab^T by assigning permutation matrices, accepting sign swaps, to each ‘axis’, translates several nice notational ideas. Symmetry or skew-symmetry in $X, Y, Z, \dots$ corresponds to commutativity or alternating, respectively. The complex numbers, for instance, have;

$$X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, Y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

And by using a more formal definition of the $\cdot$,

$$\vec{a}\vec{b} = \begin{pmatrix} \text{tr}(Xab^T) \\ \text{tr}(Yab^T) \\ \text{tr}(Zab^T) \\ \dots \end{pmatrix} = \vec{\text{tr}} \begin{pmatrix} X \\ Y \\ Z \\ \dots \end{pmatrix} (ab^T)$$

Where for the right-most statement one defines trace of a rectangular matrix as the distinct traces of the stacked square matrices, restarting the diagonal. By appending a 1 term to the input vectors, one can translate just about any quadratic map (say $x \rightarrow \vec{x}\vec{x}$) into the compact form of the above trace-product.

3 Semi-Rigid 3D Complex Mappings

Under standard theory there are no '3D Complex Numbers', however one can still hope to produce mappings, say $f(\vec{z}) = \vec{z}^2$ which are *harmonic* in each component and which persevere magnitude in the traditional sense $|\vec{z}|^2 = |z|^2$. Given this choice of quadratic, harmonic polynomials it is surprising just how far we can 'run' with these axioms and the form;

$$\vec{z}^2 = [\vec{z}^T A \vec{z}, \vec{z}^T B \vec{z}, \vec{z}^T C \vec{z}]$$

1. The condition on the Laplacian gives;

$$\text{Tr}(A) = \text{Tr}(B) = \text{Tr}(C) = 0$$

2. In order for a vector with one nonzero component to operate under the magnitude constraint;

$$[q, 0, 0]^2 = q^2 [A_{00}, B_{00}, C_{00}] \rightarrow C_{ii} = \pm \sqrt{1 - A_{ii}^2 - B_{ii}^2}$$

3. This gives a_0, a_1, b_0 (along the diagonal) as variables and gives;

$$b_1 = \frac{1}{8(a_0^2 - 1)} \left(8a_0a_1b_0 - \sqrt{(-8a_0a_1 - 4)^2 b_0^2 - 4(4a_0^2 - 4)(4a_1^2 b_0^2 - 4a_0^2 - 4a_0a_1 - 4a_1^2 - 4b_0^2 + 3) + 4b_0} \right)$$

4. By symmetry;

$$A^T = A, B^T = B, C^T = C$$

5. Suppose we take a vector with exactly two nonzero elements. We receive a set of cross terms which must equal 0;

$$a_0a_{02} + b_0b_{02} + c_0c_{02}$$

$$a_2a_{02} + b_2b_{02} + c_2c_{02}$$

etc.

6. Coupling these we find;

$$a_{01} = a_{02} = a_{12} = \sqrt{\frac{3}{4} \left(1 + \left(\frac{-a_1 + a_0 \frac{c_1}{c_0}}{b_1 - b_0 \frac{c_1}{c_0}} \right)^2 + \frac{a_0^2}{c_0^2} + \frac{b_0^2 \left(\frac{-a_1 + a_0 \frac{c_1}{c_0}}{b_1 - b_0 \frac{c_1}{c_0}} \right)^2}{c_0^2} + \frac{2a_0b_0}{c_0^2} \frac{-a_1 + a_0 \frac{c_1}{c_0}}{b_1 - b_0 \frac{c_1}{c_0}} \right)^{-1}}$$

$$b_{01} = b_{02} = b_{12} = \frac{-a_1a_{01} + a_0a_{01} \frac{c_1}{c_0}}{b_1 - b_0 \frac{c_1}{c_0}}$$

$$c_{01} = c_{02} = c_{12} = \frac{-a_0a_{01} - b_0b_{01}}{c_0}$$

Alas, despite these efforts, having three nonzero elements in the z vector produces an unavoidable error in magnitude, regardless of the choice of seed values.

$$|[q, r, s]| = \sqrt{(q^2 + r^2 + s^2)^2 + 6qrs^2 + 6qr(r+q)s}$$

Noting trivially that the undesirable cross terms vanish whenever q, r , or s equal 0, or when the vector $\vec{z}$ lies on the plane $x + y + z = 0$.

4 A Family of Nearly- π Series

In this short note we explore; (Suppose $\gamma \in \mathbf{R} \setminus \mathbf{N}$)

$$\sum_{n=1}^{\infty} \frac{\arctan(\cot(\pi\gamma n))}{a^n}$$

As it pertains to obtaining constants that lie arbitrarily close, but not equal to, π .

Under inspection of $\arctan \cot(\pi x)$, we have;

$$\sum_{n=1}^{\infty} \frac{(\text{floor}(\gamma n) - \gamma n) \pi + \frac{\pi}{2}}{a^n}$$

$$\pi \left(\sum_{n=1}^N \frac{\text{floor}(\gamma n)}{a^n} + \sum_{n=N+1}^{\infty} \frac{\text{floor}(\gamma n)}{a^n} \right) - \frac{\pi \gamma a}{(a-1)^2} + \frac{\pi}{2} \frac{1}{a-1}$$

N is selected such that the behavior of $\text{floor}(\gamma n)$ is ‘regular’. For, example if $\gamma = 2.9$, $\text{floor}(2.9 \cdot q) = 3q - 1$ for all $q \leq N = 10$. Or, $\text{floor}(7.01 \cdot q) = 7q$ holds for $q \leq N = 99$. The key observation behind this *nearly* π formulation is that the original sum is then;

$$\pi \left(\text{floor}(\gamma) C_1(N) + C_2(N) + \epsilon - \frac{\gamma a}{(a-1)^2} + \frac{1}{2} \frac{1}{a-1} \right)$$

Where $C_1(N)$ and $C_2(N)$, dependent on the truncation of the sum at N , are known exactly and where;

$$0 < \epsilon = \sum_{n=N+1}^{\infty} \frac{\text{floor}(\gamma n)}{a^n} \leq \gamma a^{-N} \frac{(a-1)N + a}{(a-1)^2}$$

Is both bounded and exceedingly small. Thus, choosing γ via rational convergents to $1/\pi$ produces a family of truly remarkable series. Stepping through the calculations for $\gamma = 22/\pi$ and $a = 22$ for instance;

$$\sum_{n=1}^{\infty} \frac{\arctan(\cot(22n))}{22^n} = \pi \left(\sum_{n=1}^{354} \frac{\text{floor}(\frac{22}{\pi} n)}{22^n} + \sum_{n=354+1}^{\infty} \frac{\text{floor}(\frac{22}{\pi} n)}{22^n} \right) - \frac{22^2}{(22-1)^2} + \frac{\pi}{2} \frac{1}{22-1}$$

$$\pi \left(7 \cdot \frac{22}{441} + \frac{1}{42} + \left[-7 \sum_{n=355}^{\infty} \frac{n}{22^n} + \sum_{n=354+1}^{\infty} \frac{\text{floor}(\frac{22}{\pi} n)}{22^n} \right] \right) = \sum_{n=1}^{\infty} \frac{\arctan(\cot(22n))}{22^n} + \frac{484}{441}$$

$$\frac{126}{47} \left(\sum_{n=1}^{\infty} \frac{\arctan(\cot(22n))}{22^n} + \frac{484}{441} \right) = \pi + \epsilon_0$$

Ultimately giving an error of;

$$\epsilon_0 \leq \pi \frac{126}{47} \left(\frac{22}{\pi} - 7 \right) \sum_{n=355}^{\infty} \frac{n}{22^n} = \pi \frac{126}{47} \left(\frac{22}{\pi} - 7 \right) \frac{22^{-354} (21 \cdot 354 + 22)}{441} \sim 2.4 \cdot 10^{-476}$$

And certainly this family of series works for any a and for any $\gamma = \frac{p}{q\pi}$ to avoid circularity. For comparison, similar calculations for $\gamma = 355/\pi$ have an N value of 104218. It should be noted that $\cot(p)$ being large for p , the numerator of a convergent for π , is to be expected trivially, and hence the composition with $\arctan$ within the summation gives immediate estimates for $\pi/2$.

5 Doubling The Cube

To double the cube requires, much like squaring the circle, the impossible straightedge-and-compass construction of $2^{1/3}$. Let's do precisely this, only up to a numerical approximation;

1. Begin with a line upon which the segment $\overline{AB}$ is made with known length 1. On this line, mark out a segment $\overline{BC}$ of length 7, followed by two swivels of the compass to give segment $\overline{CD}$ of length 14. Add segment $\overline{DE}$ of length 14 again to the end with a swivel of the compass, before turning around and marking point F on the line a length $\overline{AC}$ from the E on $\overline{DE}$. Finally, add a point G on the line segment $\overline{CD}$ such that the segment $\overline{CG}$ is of length 3.
2. From the segment of length 1, construct a perpendicular line through B and mark the point H which is a length 1 from B , giving a right triangle of side lengths 1 and 1, and giving the hypotenuse $\sqrt{2}$. From the segment of length 7 construct a right triangle of side lengths 7 and 14, conveniently housed in the neighboring segment, and collect the hypotenuse $7\sqrt{5}$. Lastly, from the segment of length 3, construct a right triangle with side lengths 3 and 2 and collect the hypotenuse $\sqrt{13}$.
3. Construct a perpendicular line off of point F , which to repeat is a distance 28 from A , and onto this place each of the gathered radicals: 1 of $\sqrt{2}$, 2 of $\sqrt{13}$ and 1 of $7\sqrt{5}$ end to end on this line, followed by the length $\overline{GD}$, which conveniently gives a length of 11. Call the end of these adjacent lengths the point R , now a distance $\sqrt{2} + 2\sqrt{13} + 7\sqrt{5} + 11$ from F and strike the line segment $\overline{RA}$.
4. The intersection of lines $\overleftrightarrow{RA}$ and $\overleftrightarrow{BH}$ forms a point, P , who's distance from B gives the side lengths of the doubled-cube.

Should the ancient Greeks have constructed an industrial wine-cube of sides $\overline{PB}$ meters, it would have a volume of $2 + 4.35...10^{-7}$ cubic meters, an error of *less than milliliter*. This geometric exercise is of course inspired by Ramanujan's famous approximate 'Squaring of the Circle' using the radical approximation technique, and who's methods, approximations, and execution is far better than anything demonstrated here.

6 Non-Convolutional Blurring

Following standard Total Variation Denoising (where $|\cdot|$ indicates the Frobenius Norm), given an image matrix Y we can seek to minimize the following functional of X to balance both a fidelity term and a variational term with a weight ratio λ .

$$|X - Y|^2 + \lambda |AXB - X|^2$$

One convenient value for these matrices is simply a shift;

$$A = B = \begin{pmatrix} 0 & 0 & 0... & 1 \\ 1 & 0 & 0... & 0 \\ 0 & 1 & 0... & 0 \\ 0 & 0 & 1... & 0 \end{pmatrix}$$

Which in essence compute the gradient of X . For a fixed λ , the larger the less the 'preference' to the fidelity term, we compute the gradient to be;

$$\nabla_X = 2(X - Y) + 2\lambda [X - AXB - A^T X B^T + A^T A X B B^T]$$

This can be easily consolidated if either A or B are the identity. Let's make the dubious calculation that we compound the transformations corresponding to $A = I_N$ and $B = I_N$ simultaneously;

$$X = (I + \lambda [I - A - A^T - A^T A])^{-1} Y (I + \lambda [I - B - B^T - B B^T])^{-1}$$

The effect of this transformation is blurring without convolution of varying kernels A , B , and denoising strength λ . Assuming the shift matrices discussed previously are used, we receive blurring matrices

that are nearly symmetric tri-diagonal, with patterned inverses. And, as λ tends to infinity, the blurring matrices approach the matrices of all ones divided by N , giving a complete averaging effect for *minimal* variation.

6.1 Remarks on Feature Detection

Suppose we take a standard blurred representation of an image Y and an energy denoised version of Y , as computed above, with similar variations, as computed by the $|AYB - Y|^2$ term in the defining metric. The difference between the two, perhaps in log form, has contours which represent the *highest-energy features 'in-focus'*.

$$[\text{Blur}\{Y\} - X] \text{ or } \ln |\text{Blur}\{Y\} - X|$$

By edge-detecting on the difference, or log of the difference, between the blur effects detailed above, one can obtain the most important regions of detail to the *variation* of the pixels at a certain 'resolution'.